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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Classification of Malware Behavior Based on Known Malware Samples

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Members

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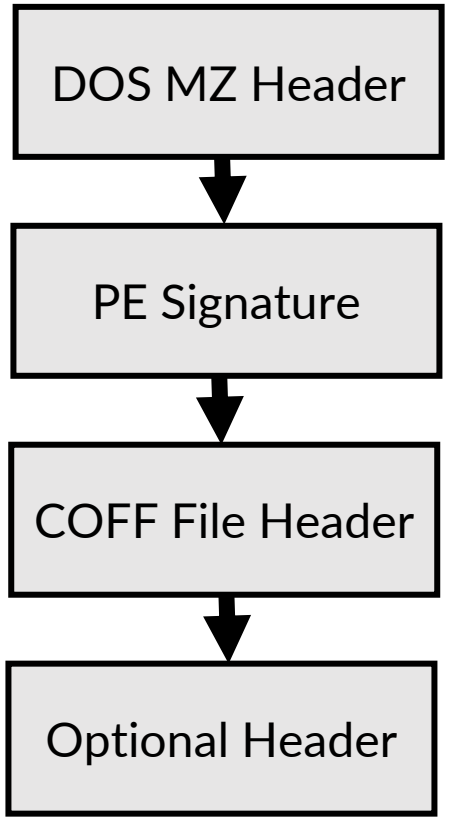
1. Theoretical Background – PE Format & Disassembly

- **Headers:**



Responsible for backward compatibility

Indicates the beginning of NT file header

Specify machine types and number of sections

Contains entry point virtual address, image base, and directories

1. Theoretical Background – PE Format & Disassembly

- **Sections:**
  - .text or .code: Contains executable machine instructions
  - .data: Holds initialized global variables.
  - .rdata: Contains read-only variables, debug directories, and the Import Address Table (IAT).
  - .rsrc: Contains application resources (icons, menus, versions).

1. Theoretical Background – PE Format & Diassembly

- **Import Address Table (IAT):** Records the dynamic link libraries (DLLs) and functions imported from the system API (***LoadLibrary***, ***GetProcAddress***, ...)

- **Disassembly:** Reversing raw binary bytes in ***.text*** into x86 assembly mnemonics (***mov***, ***xor***, ***cmp***, ***etc***.) for opcode analysis.

1. Theoretical Background – Shannon Entropy

- For a discrete random variable *X* representing byte values (from 0x00 to 0xFF) in a binary file or section, the Shannon entropy *H*(*X*) is computed as:

$$H(X) = -\sum_{i=1}^{n} P(x_i) log P(x_i)$$

While n = 256 possible bytes values, $P(x_i)$ is the empirical probability of occurrence of byte x_i inside an analyzed block

- **Low entropy (0.0 – 5.0):** Highly structured, predictable data
- **Medium entropy (5.0 – 6.8):** Normal executable code or compiler output.
- **High entropy (7.0 – 8.0):** Indicates high randomness, approaching uniform distribution

→ This is typical of encrypted data, compressed buffers, or packing payloads, since cryptography and compression algorithms aim to eliminate statistical patterns.

1. Theoretical Background – Packing & Obfuscation

- **W^X Permission Violation:** Sections that are simultaneously Writable and Executable. Standard compilers separate these, but packers break this rule to decrypt code directly into memory.
- **Virtual vs. Raw Size Mismatch:** When the virtual size of a section in memory is significantly larger than its size on disk. This points to an expansion container for an unpacked payload.
- **Low/Zero Import Counts:** Packed files often hide their IAT, importing only baseline functions like ***LoadLibraryA*** and ***GetProcAddress*** to dynamically load APIs at runtime.
- **Suspicious Section Names:** Section headers named ***.upx0***, ***.aspack***, or containing unexpected characters like **/.**

1. Theoretical Background – Behavioral FSM: API Chain Tracking

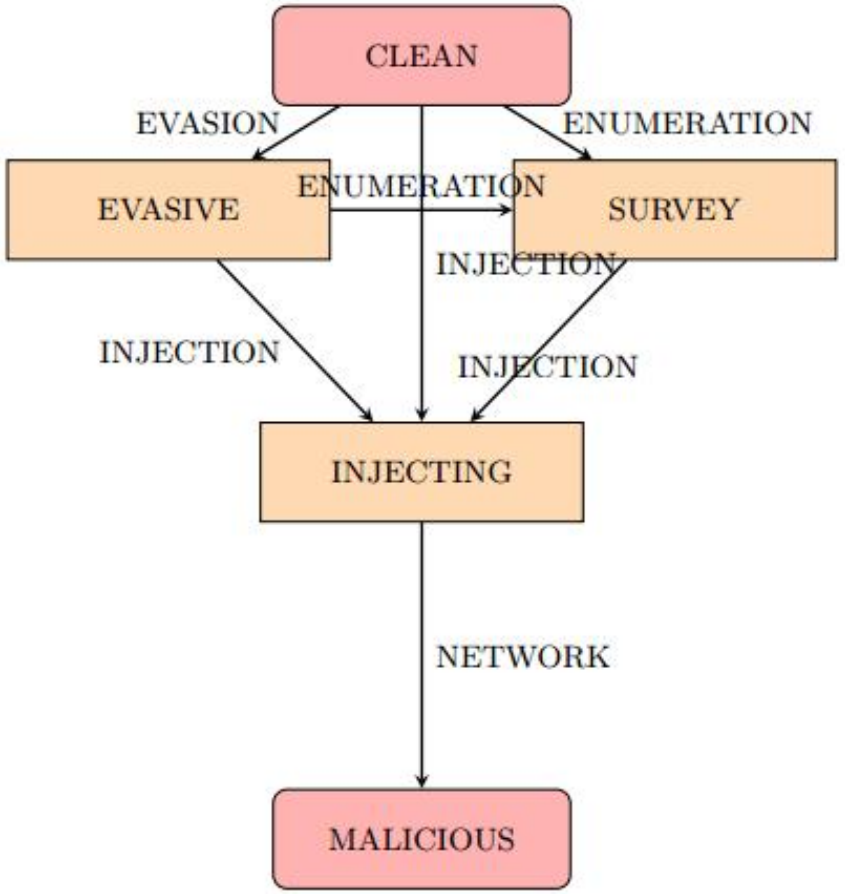


Figure 1: Behavioral FSM State Transitions

Score Impact:

CLEAN / EVASIVE: +0 pts

SURVEY: +5 pts

INJECTING: +15 pts

MALICIOUS: +35 pts

1. Theoretical Background – Decryption Loop FSM

- **State 1: START** → Scans disassembled assembly opcodes. If a decryption operation *(xor, rol, ror)* is detected, transition to **XOR_FOUND**.
- **State 2: XOR_FOUND** → If the subsequent instruction updates loop pointers *(cmp, test, dec, inc, add, sub)*, transition to **LOOP_FOUND**. Otherwise, reset.
- **State 3: LOOP_FOUND** → If the next instruction is a conditional jump or loop back *(jnz, jne, loop, jmp)*, transition to **DECRYPTOR**. Otherwise, reset.
- **Result:** Reaching the **DECRYPTOR** state signals a decryption stub, adding **+25** points to the heuristic score.

1. Theoretical Background – Weighted Heuristic Scoring Model

Table 2: Heuristic Scoring Weights and Indicators

Category	Anomalous Indicator	Score Weight
Code Section	Missing standard code section (`.text` renamed/packed)	+35
Entropy	Section entropy > 7.9 (encrypted/packed)	+45
	Section entropy > 7.5	+30
	Section entropy > 7.1	+15
Section Permissions	W^X violation (Writable + Executable section)	+35
Section Sizes	Virtual size vs Raw size mismatch (likely packed container)	+25
Section Naming	Known packed section name (e.g. `.upx0`, `.aspack`)	+35
	Section name starts with `/` (compiler anomaly/obfuscated)	+35
Overlay Details	Overlay size > 3 MB and file size > 4 MB	+55
	Overlay size > 150 KB and file size > 500 KB	+30
Opcode Density	Low opcode density (< 300 instructions in executable file)	+25
Imports Structure	Zero imports (extreme packers / shellcode)	+60
	Fewer than 5 imported functions	+30
Kernel Drivers	Kernel driver containing process control APIs	+50
Debug & Evasion	Advanced anti-debugging or debugging APIs	+30
Process Injection	Thread manipulation: Imports `SetThreadContext`	+20
	Dynamic API Resolution (`LoadLibrary` + `GetProcAddress` + `VirtualAlloc`)	+15
Instruction Ratio	Extremely high `add` opcode ratio (> 40%)	+40
	High `add` opcode ratio (> 30%)	+25
Strings	High-risk strings (commands targeting backups/AV exclusions)	+30

Dynamic Discounts (FP Mitigation):	
Export count > 50	-35
High DLL import count (>12)	-20
Windows API-Set DLL imports (> 4 ***api-ms-win**** libraries)	-20

➡ **Classification Threshold**: Files with a score ≥ 50 are classified as **MALWARE**.

1. Theoretical Background – Heuristic Probability Curve

To express threat likelihood as a percentage rather than a raw score, the engine computes a Malware Probability *P* using an exponential curve:

$$P = \begin{cases} 0.0 & \text{if } S \leq 0 \\ \left(\frac{S}{\text{Threshold}}\right) \times 50.0 & \text{if } 0 < S < \text{Threshold} \\ 50.0 + 50.0 \times \left(1 - e^{-0.05 \times (S - \text{Threshold})}\right) & \text{if } S \geq \text{Threshold} \end{cases}$$

While S is the heuristic score, Threshold = 50

➡ Scores that are far above the threshold asymptotically approach 100% malware probability.

2. System Architecture

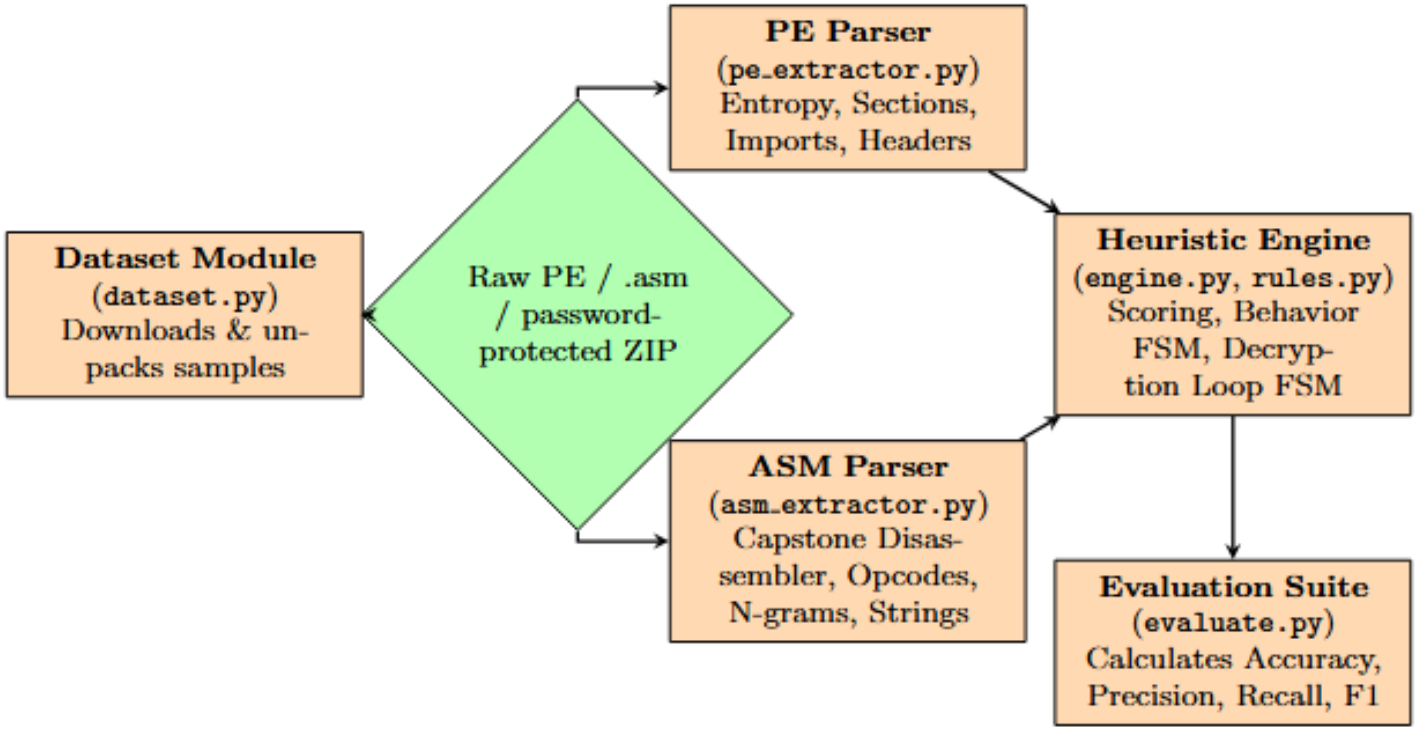


Figure 2: Malware Detection Architecture Flow

3. Evaluation

The performance of the heuristic engine was validated against a large-scale static features database containing 1,797 samples:

- **Benign Samples**: 993 legitimate Windows executables, DLLs, and system drivers.
- **Malware Samples**: 804 wild malware binaries collected securely from MalwareBazaar.

Metric	Optimized Heuristic Model
Accuracy	95.10%
Precision	94.53%
Recall (Sensitivity)	94.53%
F1-Score	94.53%

Table 4: Optimized Malware Detector Confusion Matrix ($N = 1,797$)

	Actual Malware	**Actual Benign**
Predicted Malware	760 (True Positive)	44 (False Positive)
Predicted Benign	44 (False Negative)	949 (True Negative)

4. Conclusion

- **Core strength**:
  - **Safety & Speed:** Evaluates files statically in milliseconds with zero threat execution risk.
  - **Robust Obfuscation Heuristics:** Catches packer stubs using entropy triggers and a custom 3-gram assembly FSM.

- **Drawbacks:**
  - **API Obfuscation Blindspot:** Unable to map API calls if resolved dynamically using custom hashes.
  - **Payload Encryption Limits:** Static analysis cannot parse hidden instructions before they are loaded into RAM.
  - **Dual-Use Tool Overlaps:** Legitimate debuggers and command engines naturally trigger injection rule

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THANK YOU !